

# Methodological Limitations and Remedies

Drafted to pre-empt the methodological concerns raised in peer review (self-report bias, social desirability, common method variance, construct discriminant validity, ceiling effects, priming, and generalizability). Intended for the Methodology chapter (design-stage remedies) and the Limitations section (residual constraints). Anticipating these in writing converts predictable committee criticism into evidence of methodological awareness.

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## 1. Self-report measurement and construct interpretation

All constructs are measured through auditor self-report. Because anchoring operates partly outside conscious awareness, self-reported items capture **perceived** practices, conditions, and judgment tendencies rather than directly observed behavior. This is a deliberate and defensible scope decision: the study validates a *measurement model* of the organizational and cognitive conditions auditors associate with anchoring reduction, consistent with the survey tradition in audit judgment-and-decision-making research, rather than experimentally measuring anchoring magnitude. The constructs are therefore interpreted as **perceptions of process and judgment conditions**, and this framing is stated explicitly wherever the results are discussed. An experimental or archival follow-up that observes anchoring behavior directly is identified as a natural extension.

## 2. Common method variance (the primary methodological risk)

Because the independent variables, mediators, and dependent variable are collected from a single respondent, in one session, on a common response format, the design is exposed to common method variance (CMV). Following the procedural and statistical framework of Podsakoff, MacKenzie, Lee, and Podsakoff (2003), the study incorporates **procedural remedies at the design stage** and **statistical diagnostics at the analysis stage**:

### Procedural remedies (built into the instrument):

- **Respondent anonymity** with no identifying information collected, reducing evaluation apprehension and the incentive to respond in a self-serving or consistent manner.
- **Reverse-coded items** within every construct, disrupting response-pattern consistency.
- **Block randomization** of the intervention constructs, so the predictor blocks are psychologically separated and not encountered in a fixed, theory-revealing order.
- **Varied item framing and an open-ended item**, reducing reliance on a single cognitive response mode.

### Statistical diagnostics (planned for analysis):

- **Harman's single-factor test** — examine whether a single unrotated factor accounts for the majority of covariance among items (a substantial single factor would signal CMV).
- **Common-latent-factor / marker-variable assessment** — compare construct loadings with and without a method factor to estimate method-attributable variance, per Podsakoff et al. (2003).

Residual CMV that cannot be ruled out is acknowledged as a limitation.

## 3. Discriminant validity among conceptually related constructs

Several constructs are conceptually adjacent — Structured Auditing Processes, Audit Process Rigor, Independent Reviews, and Regulatory and Professional Guidance all describe organizational controls, and Auditor Judgment Quality overlaps in surface content with Reduction in Anchoring Bias. **This overlap is not a flaw to be argued away; it is precisely the empirical question the exploratory factor analysis is designed to answer.** The EFA pattern matrix tests whether items load on distinct factors as theorized. The reporting plan makes discriminant validity falsifiable:

- Inspect the rotated pattern matrix for **cross-loadings** (items loading  $\geq .32$  on more than one factor are flagged), and examine **inter-factor correlations**.
- For retained scales, assess discriminant validity via the **Fornell and Larcker (1981) criterion** — the average variance extracted (AVE) for each construct should exceed its shared variance with other constructs.
- **Pre-committed decision rule:** if two constructs fail to separate empirically, that is a reported finding; the affected items are revised or the constructs are merged, with advisor sign-off, rather than forced apart.

This turns the reviewer's concern into a stated, testable hypothesis about the factor structure.

#### 4. Social desirability

Some items describe behaviors auditors are professionally expected to exhibit (objectivity, skepticism, independent evidence use), inviting socially desirable responding. The primary procedural mitigation is **anonymity** with no identifying data, which removes the audience that motivates impression management. Reverse-coded items and the neutral, non-evaluative framing of the instructions further reduce the pull toward a "correct" answer. Residual social-desirability bias is acknowledged in the limitations; a social-desirability marker is identified as an option for a confirmatory follow-up.

#### 5. Ceiling effects and response distribution

Positively framed items answered by experienced professionals may compress toward the agreement end, reducing variance. Mitigations include the **reverse-coded items**, a **five-point scale that retains a neutral midpoint**, and the examination of **item-level means, variances, and skewness** during data screening. Items exhibiting severe ceiling effects (minimal variance) are flagged and considered for removal during the EFA, where low communalities and weak loadings would also surface them.

#### 6. Priming and topic salience

A brief topic orientation was added at reviewer request to aid comprehension; a second reviewer noted that naming the bias could prime responses. To balance the two, the orientation is kept **neutral and general** — **describing professional judgment in recurring engagements without instructing respondents on how anchoring "should" be avoided.** The trade-off is documented: the orientation favors shared comprehension over maximal naïveté, an acceptable choice for a transparent validation survey rather than a deception experiment. Residual priming is noted as a limitation.

#### 7. Generalizability across audit environments

Some items presuppose formal rotation policies, advanced analytics, engagement-quality review, or regulatory inspection, which may not apply uniformly to internal, governmental, or small-firm auditors. Item

wording was broadened during review to include internal-audit terminology (e.g., "auditees," IIA/COSO). The demographic block captures **audit specialization (internal/external), years of experience by type, organization type, firm size, industry, long-term-engagement exposure, and credentials**, enabling these factors to serve as **controls and moderators** and allowing sensitivity analyses across subgroups. Generalization beyond the achieved sample's composition is bounded accordingly in the limitations.

## 8. Survey length and respondent fatigue

The instrument's length (55 construct items plus screeners, controls, and an open item) creates fatigue risk that can produce straight-lining toward the end. Mitigations include the **five-point (lower-burden) scale, block randomization** so any end-of-survey fatigue is distributed across different constructs rather than concentrated on one, a **progress indicator**, and the **embedded attention checks** plus **page-timing metadata** used to identify and exclude careless responders during data cleaning.

### Reviewer-comment disposition (audit trail)

| #  | Reviewer concern                   | Where addressed                                                                           |
|----|------------------------------------|-------------------------------------------------------------------------------------------|
| 1  | Self-report reliance               | §1 + Limitations                                                                          |
| 2  | Social desirability                | §4 (anonymity) + Limitations                                                              |
| 3  | Construct overlap (SAP/APR/IR/RPG) | §3 — EFA discriminant validity (Fornell & Larcker)                                        |
| 4  | IV/DV overlap (AJQ/RAB)            | §3 — same                                                                                 |
| 5  | Priming from definition            | §6 — orientation trimmed to neutral                                                       |
| 6  | Common method bias                 | §2 — procedural + statistical remedies (Podsakoff et al.)                                 |
| 7  | Ceiling effects                    | §5                                                                                        |
| 8  | More reverse items                 | 1 per construct retained; acknowledged (added reverse items are a future-revision option) |
| 9  | Applicability across environments  | §7 + demographic controls                                                                 |
| 10 | Missing demographics               | Already in instrument (specialization, experience, firm size, region, certs)              |
| 11 | Open-ended timeframe               | Instrument tweak — add "in the past 12 months"                                            |
| 12 | Attention checks                   | Already in instrument (two embedded)                                                      |
| 13 | Length / fatigue                   | §8                                                                                        |

### New verified references (add to the pool)

- Fornell, C., & Larcker, D. F. (1981). Evaluating structural equation models with unobservable variables and measurement error. *Journal of Marketing Research*, 18(1), 39–50.

- Podsakoff, P. M., MacKenzie, S. B., Lee, J. Y., & Podsakoff, N. P. (2003). Common method biases in behavioral research: A critical review of the literature and recommended remedies. *Journal of Applied Psychology*, 88(5), 879–903.

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